Risk Assessment & Mitigation

Risk Management Plan

The project team will identify, assess, and monitor risks throughout development. Each risk will be evaluated based on its likelihood and potential impact.

Risk	Likelihood	Impact	Mitigation Strategy
Unexpected AWS costs	Medium	High	Set a project budget, use AWS Budgets and billing alerts, choose appropriately sized resources, stop unused instances, and review storage costs regularly.
Difficulty integrating Computer Vision, backend, and frontend components	High	High	Define API contracts early, use modular architecture, agree on event schemas, and integrate components incrementally.
Low detection accuracy in poor lighting or crowded scenes	Medium	High	Test representative videos, tune confidence thresholds, evaluate a held-out dataset, and document environmental limitations.
Tracking ID switches when people overlap or disappear	High	Medium–High	Evaluate tracking algorithms, tune tracker parameters, measure ID switches, and avoid treating temporary tracking IDs as verified worker identities.
Insufficient or unsuitable video data	Medium	High	Use authorized sample videos, prepare alternative test clips, annotate a small evaluation dataset, and define acceptance criteria before final testing.
Slow processing or high latency	Medium	High	Benchmark the pipeline, resize frames when appropriate, sample frames carefully, optimize

			inference, and select suitable compute resources.
AWS deployment or permissions failures	Medium	High	Test deployment early, follow least-privilege IAM practices, validate service connectivity, and keep a documented deployment procedure.
Database or evidence storage failure	Low–Medium	High	Implement error handling, verify successful uploads, monitor failures, configure suitable backups, and test recovery procedures.
Camera stream disconnection	Medium	Medium–High	Implement health checks, detect missing frames, log connection failures, and attempt controlled reconnection.
Excessive false alerts	Medium–High	High	Tune event thresholds, add dwell-time conditions, evaluate false alerts per hour, and provide human review for important incidents.
Privacy or unauthorized video access	Medium	High	Use authorized footage, restrict access through IAM and application roles, secure evidence links, define retention policies, and maintain audit logs.
Scope expansion and schedule delays	Medium	Medium–High	Prioritize the single-camera MVP, track changes, reserve time for integration, and defer nonessential features.

Risk Monitoring and Response

The team will review the risk register during weekly progress meetings. High-impact risks will be prioritized, assigned to an owner, and tracked until resolved or reduced to an acceptable level.

AWS spending will be reviewed throughout development rather than only at project completion. The team will also maintain a working local development environment to reduce the impact of cloud deployment issues.

If a risk cannot be mitigated within the planned schedule, the team may reduce the feature scope while preserving the core requirements: person detection, tracking, rule-based event generation, incident review, and a working demonstration.